

Sidekick-mini usage analysis

Student progress export — 2026-04-23

Analysis of **23,880 step responses** from **215 students** across **6 mini-lessons** (14 questions / 142 steps), after filtering out preview/archived rows.

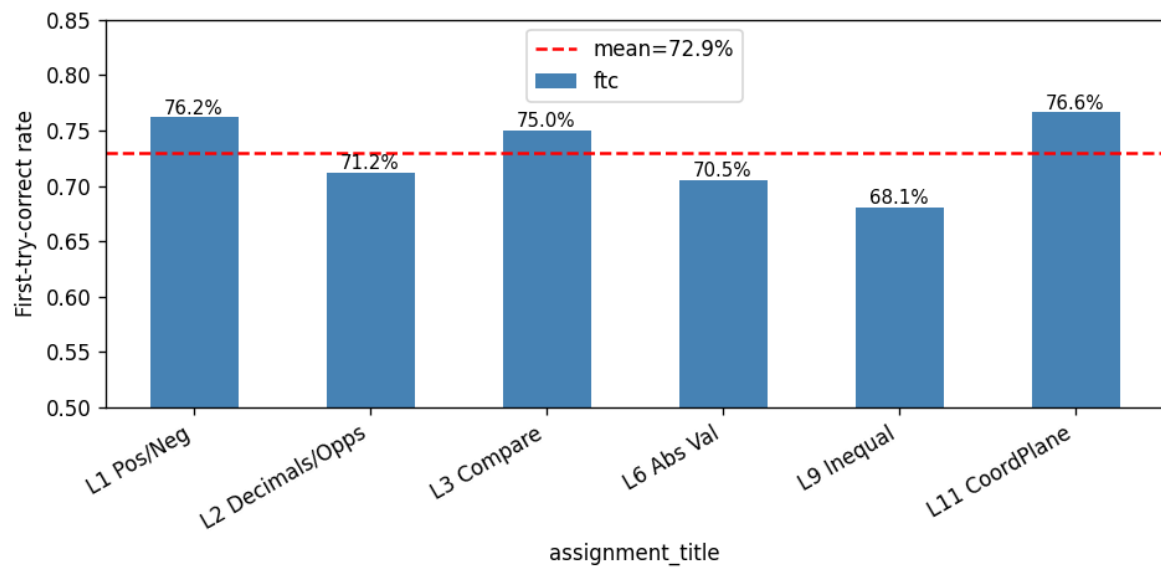
Headline numbers

Metric	Value
Overall first-try-correct (FTC)	73.2%
Average attempts per step	1.64
Step-level replay rate	25.7%
Unique students	215
Unique mini-lessons	6
Unique questions (KCs)	14
Unique steps	142

Lesson ranking (hardest → easiest)

Lesson	FTC	Avg attempts	Replay rate
L9 Intro to Inequalities	68.1%	1.92	30.6%
L6 Absolute Value	70.5%	1.39	27.6%
L2 Points on the Number Line	71.2%	1.94	27.2%
L3 Comparing +/-	75.0%	1.47	24.0%
L1 Positive and Negative	76.2%	1.41	23.3%
L11 Coordinate Plane	76.6%	1.67	22.5%

L9 (Inequalities) is the clear outlier — hardest by every metric and the only lesson with replay rate above 30%.



Conceptually hardest steps

Steps where students hit a wall — low first-try-correct plus elevated attempts across most of the class, not just a tail.

Lesson / KC	Step	FTC	Avg att.
L9 Graphing inequalities	step-7	20.8%	5.17
L9 Graphing inequalities	step-9	31.8%	3.69
L9 Inequality solutions	step-8	34.4%	3.96
L6 Absolute Value	step-3	34.4%	1.67
L1 Negative numbers	step-7	36.2%	2.61
L11 Finding points on grid	step-1	41.1%	2.93
L2 Opposites	step-5	42.2%	2.95
L6 Absolute Value	step-9	43.3%	1.57
L2 Decimals	step-0	44.1%	4.48
L1 Comparing integers	step-6	44.5%	2.37
L3 > and < symbols	step-8	46.0%	3.12

Likely UI / input friction

Steps where max attempts explodes despite moderate-to-high FTC. p90 stays small — this is individual sessions stuck in a broken state, not broad failure. Worth replaying these sessions to see what the interaction did.

Step	FTC	Avg	p90	Max attempts
L11 Finding points on grid · step-4	77.9%	3.45	2.0	199
L2 Decimals · step-1	71.6%	2.68	4.0	121
L11 Distance planes · step-1	79.7%	2.68	2.0	90
L2 Decimals · step-3	55.7%	3.74	7.0	88
L3 >/< symbols · step-8	46.0%	3.12	5.0	81
L9 Graphing ineq. · step-4	93.6%	1.55	1.0	81
L3 Comparing integers · step-5	62.7%	3.65	8.0	74
L2 Decimals · step-0	44.1%	4.48	11.0	71

Q1. How exactly did I verify that students finish all steps?

Each response_id is one student's single play-through of one question, with one row per step_key (typically 10 rows, 0–9). Counting distinct response_ids at each step within a question, the count is identical for step-0 and step-9 in every question. So **once a student opens a question, they reach all of its steps** — no mid-question drop-off.

But that says nothing about whether students complete every question in a mini-lesson. The true cross-question completion rates:

Mini-lesson	Questions	Students	Completed all Qs
L1 Positive/Negative	3	213	98.6%
L2 Points on NL	2	209	97.6%
L3 Comparing	2	203	98.5%
L6 Absolute Value	1	195	100%
L9 Inequalities	3	173	75.7% ■
L11 Coord Plane	3	95	72.6% ■

Honest version: within a question, no mid-question drop-off. Between questions in L9 and L11, about a quarter of students never started the later questions — consistent with L9 being the hardest lesson and L11 being the most recently released.

Q2. Can we see end-of-lesson reassessment results?

Partially. After re-examining the schema, first_try_correct and replayed are near-perfectly anti-correlated ($r = -0.999$):

Case	Rows	Interpretation
ftc=1, replayed=0	17,447	Right first try; no replay needed (attempts=1)
ftc=0, replayed=1	6,159	Wrong first try; re-asked at end of lesson
ftc=0, replayed=0	269	Session likely ended before replay
ftc=1, replayed=1	5	Edge cases

So replayed=t marks the rows where a student missed first time and got the end-of-lesson reassessment. attempts counts total tries across all phases. Every step eventually gets answered (we saw no mid-question drop-off), so all 6,159 replayed rows were eventually answered correctly. The question is how painfully:

Total attempts on missed-then-replayed steps	Count	Share
2 (wrong once, right once)	4,334	70.3%
3	651	10.6%
4	328	5.3%
5–9	583	9.5%
10+	263	4.3%

Take: of the students who got a step wrong on first try, about **70% turned it around with one additional correct attempt** (most likely the end-of-lesson reassessment). About 15% needed 4+ total tries. Strong evidence that the reassess-at-end mechanic is doing real work.

Caveat: I can't distinguish the immediate retry from the end-of-lesson retry — both are lumped into attempts. To separate them you'd need the backend event log.

Q3. Explaining “max attempts”

attempts is the total number of tries a single student made on one step. max_attempts is the maximum across all students who did that step. The key diagnostic is the gap between p90 and max:

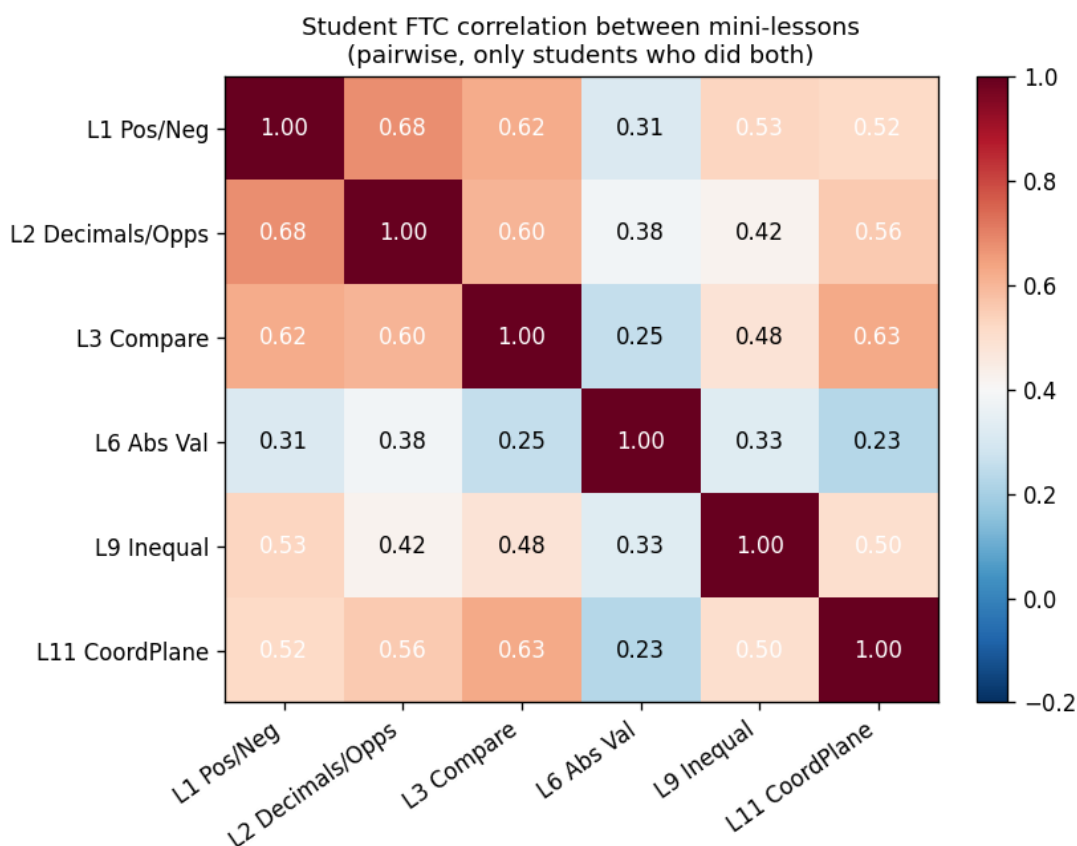
Step	Avg	p90	Max	Diagnostic
L11 Finding points · step-4	3.45	2	199	90% did it in ≤ 2 . One student: 199. Almost certainly a UI lockup or mass-click.
L2 Decimals · step-1	2.68	4	121	Noticeable tail AND one outlier — UI friction plus a few stuck students.
L2 Decimals · step-0	4.48	11	71	p90=11: many students took ≥ 10 tries. Conceptually hard, not just one outlier.
L9 Graphing · step-4	1.55	1	81	90% first try. One student: 81. Pure outlier — browser/input glitch.

So “199 max attempts on coord plane step-4” doesn't mean students fail it broadly (77.9% get it first try). It means **one individual spent ages clicking** — strong smell of an unresponsive interaction or a validator rejecting valid-looking answers. Worth logging clicks on those steps and replaying sessions.

Where avg + p90 + max are all high (L2 Decimals step-0: 4.48 / 11 / 71), that's the worst case: conceptually hard **and** UI-friction-y.

Q4. Correlation between lessons

Student-level first-try-correct across pairs of mini-lessons. Pairwise: only students who did both lessons.



Most pairs correlate 0.5–0.7 — the expected “some kids are stronger overall” pattern. **L6 Absolute Value is the exception: 0.23–0.38 with every other lesson.** That’s meaningfully lower — see Q6.

Q5. Top-offender steps: content + hypotheses

L9 “Graphing inequalities” step-7 — 21% FTC, avg 5.17 attempts

“The reactor will melt down if the core temperature rises above 3° . Write an inequality for that. Choose a letter to represent the temperature.”

UI: [type variable] [dropdown] [type number]. Target: $t > 3$ but any variable accepted.

Hypothesis: This is the *first* step that asks students to pick their own variable name. Nothing earlier taught them “any letter works” for the variable box. They likely type nothing / “x” / “temperature” and hit errors. The answer space is combinatorial (variable $\times$ symbol $\times$ number), so noise is high.

Fix: either pre-fill the variable (e.g. “t”) or add an inline tutorial on the first open-variable step. Hint should spell out: “Type any letter — ‘t’ works well for temperature.”

L9 “Graphing” step-9 — 32% FTC

“A sign on Math Highway reads ‘ $s < 65$ mph’. What do you think it means?”

MC: “Your speed must be less than 65 mph” vs “more than 65 mph”.

Hypothesis: On a 2-choice MC, 32% FTC is worse than random. Students are systematically reading $<$ as “greater than.” That’s a direction-reversal error — likely because in the preceding screens students were *building* inequalities by picking an arrow direction; here they have to *read* a symbol cold, and nothing in the prior steps drills “ $<$ means less.”

Fix: add an interpretation step right before this one: “Remember, the open side of $<$ faces the bigger number.”

L9 “Inequality solutions” step-8 — 34% FTC, avg 3.96 attempts

“Write an inequality for which 7, 15, and 33 are solutions.”

UI: type variable + pick symbol + type number. Target: $n > 6$ or equivalent.

Hypothesis: First fully open-ended “construct-an-inequality” task. Students have been picking from choices or filling one slot — this demands three correct decisions with no scaffolding. Many-valid-answers validator can also be a trap (does $x > 5$ work? $n \geq 7$?).

Fix: break into two steps — first “pick $>$ or $<$ ”, then “pick a cutoff number that’s smaller than 7.” Or stage with the variable pre-filled.

L6 “Absolute value” step-3 — 34% FTC

“What is the absolute value of 4?”

MC: 4 (correct) vs -4 .

Hypothesis: The weirdest result in the dataset. Step-2 just before it asks “absolute value of -4 ” (answer: 4) and gets $\sim 72\%$ FTC. Then step-3 asks “absolute value of 4” and 66% of students pick -4 . They’ve overgeneralized the rule they just “learned” from step-2 — “absolute value flips the sign” — and now confidently flip positive to negative.

Fix: after step-2, insert an explicit framing: “Absolute value is always the *distance*, so it’s never negative” BEFORE the $|4| = ?$ item. Textbook case of a worked example teaching a wrong rule.

L2 “Decimals on the number line” step-0 — 44% FTC, avg 4.48, max 71

“Here is 2.5. Plot 2.1. (Each tick is a tenth: 0.1.)”

Range -3 to 3 , tickStep 0.1, labelStep 1 (only integers labeled).

Hypothesis: Two compounding issues. (1) With tickStep 0.1 on a range of 6, there are 60 ticks; at phone/Chromebook widths they may be near-invisible or too close to click precisely. (2) Students see “2.5” labeled and have to count backwards by tenths without a “2” label anywhere nearby. Many probably put 2.1 somewhere near 2.5.

Fix: add labels at each unit visible on screen AND give the reference “2” a label, not just 2.5. Consider a click-snap to nearest tenth.

L1 “Negative numbers” step-7 — 36% FTC, avg 2.61

“Add 2 to this point. Move it to where it should be.”

Type: move-point. Starts at -3 (carried from step-6), target -1 .

Hypothesis: First time in the product the interaction is “move an existing point by an arithmetic operation” rather than “plot a new point.” Students may (a) try to add a new point instead of dragging the existing one, (b) move it left (subtract) because of “adding a negative to a negative”-style confusion.

Fix: change the instruction to make the interaction mode explicit: “**Drag** this point 2 to the right (adding 2).” The word “move” underplays that this is a new interaction type.

Q6. Why L6 tests something its prerequisites don't prepare for

L6's concept graph looks right on paper: negatives (L1) → number-line comparison (L2, L3) → absolute value (L6). But the actual step flow reveals a mismatch.

Everything before L6 teaches: signs matter, direction matters, bigger-negative-is-smaller. L6 then asks students to **ignore the sign** and answer about *distance*. That's the opposite mental move.

The two cliff steps:

Step-3: “ $|4| = ?$ ” (34% FTC). The prerequisite lessons have drilled “negatives flip to positives and vice versa” (L2 Opposites: “What's the opposite of -4 ?” → 4). Step-2 of L6 then asks “ $|-4| = ?$ ” (answer 4) — which *looks identical* to an “opposites” problem from L2. So when step-3 asks “ $|4|$?”, students apply the L2 opposites rule and answer -4 . The lesson has taught absolute value via an example that's indistinguishable from “take the opposite.”

Step-9: “opposites can have different absolute values — agree/disagree?” (43% FTC). The negative framing (“can... can never”) combined with the concept-bridge back to L2 opposites means students have to hold two ideas at once (opposites = mirror image, absolute value = distance) and see they're compatible. Nothing earlier demands that synthesis.

So the low correlation with L1/L2/L3 ($r \approx 0.25-0.38$) is real: students who mastered “plot -4 ,” “is -5 less than -3 ?”, “opposite of 7?” are not necessarily the ones who grasp “distance from zero is always positive.” L6 tests a **conceptually orthogonal** skill (magnitude vs signed value) that the prerequisites never directly set up.

Fix:

1. Insert a dedicated bridging mini-lesson: “how far is each number from 0?” (using only positive answers, no sign talk) *before* introducing the word “absolute value.”
2. Re-sequence step-3 to come *before* step-2, so students meet “ $|4|=4$ ” (trivial) before “ $|-4|=4$ ” (the interesting case), not the reverse.

Analysis based on sidekick-mini-progress-2026-04-23.csv (24,053 rows / 23,880 after preview filter).